

7. [Maximum mark: 5]

Consider the complex number $z = 3^{i-1}$.

(a) Write the integer 3 in the form e^a where $a \in \mathbb{R}$. [1]

(b) Hence, giving your answers in the form $p \cos(\ln q)$ where $p, q \in \mathbb{Q}^+$, find

(i) $\operatorname{Re}(z)$;

(ii) $\operatorname{Re}\left(\frac{1}{z}\right)$. [4]

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